

TASK 21:ASSIGNMENT NO 21

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GitHub Repository Link: <https://github.com/anujaw193-star/python-project-collection>

1. (Setup and Libraries)

```
# streamlit run app.py
import streamlit as st
import joblib
import pandas as pd
```

2. (Loading Model and Preprocessing Objects)

```
# Model Fetching
joblib.load("./models/model.pkl")
joblib.load("./models/columns.pkl")
joblib.load("./models/scaler.pkl")
```

3. (Page Configuration)

```
st.set_page_config(
    page_title= "Ford Car Price Prediction",
    layout="centered"
```

Output:



Ford Car Price Prediction

4. (Title and Description)

```
# Display the main title of the application
st.title("Ford Car Price Prediction")
# Display a short description for the user
st.write("Enter the car details below to predict its selling price.")
```

Output:

Ford Car Price Prediction

Enter the car details below to predict its selling price.

5. (Numerical Input Fields)

```
app.py > ...
1  # streamlit run app.py
2  import streamlit as st
3  import joblib
4  import pandas as pd
5  # Model Fetching
6  joblib.load("./models/model.pkl")
7  joblib.load("./models/columns.pkl")
8  joblib.load("./models/scaler.pkl")
9  # UI
10
11  st.set_page_config(
12      page_title= "Ford Car Price Prediction",
13      layout="centered"
14  )
15
16  # Display the main title of the application
17  st.title("Ford Car Price Prediction")
18  # Display a short description for the user
19  st.write("Enter the car details below to predict its selling price.")
20  # Numerical input fields
21
22  # Manufacturing Year
23  year = st.number_input(
24      "Manufacturing Year",
25      min_value=1990,
26      max_value=2025,
27      value=2018
28  )
29
30  # Mileage
31  mileage = st.number_input(
32      "Mileage",
33      min_value=0,
34      max_value=500000,
35      value=30000
36  )
```

```

30 # Mileage
31 mileage = st.number_input(
32     "Mileage",
33     min_value=0,
34     max_value=500000,
35     value=30000
36 )
37
38 # Road Tax
39 tax = st.number_input(
40     "Road Tax",
41     min_value=0,
42     max_value=1000,
43     value=150
44 )
45
46 # Miles Per Gallon (MPG)
47 mpg = st.number_input(
48     "MPG",
49     min_value=0.0,
50     max_value=100.0,
51     value=55.0
52 )
53
54 # Engine Size
55 engineSize = st.number_input(
56     "Engine Size (L)",
57     min_value=0.0,
58     max_value=10.0,
59     value=1.5
60 )
61
62 # Input should be feed to the model

```

Output:

Ford Car Price Prediction

Enter the car details below to predict its selling price.

Manufacturing Year

2018

Mileage

30000

Road Tax

150

MPG

55.00

Engine Size (L)

1.50

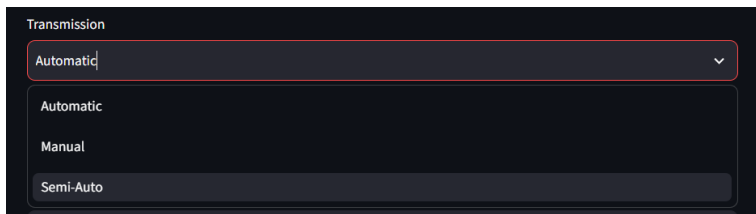
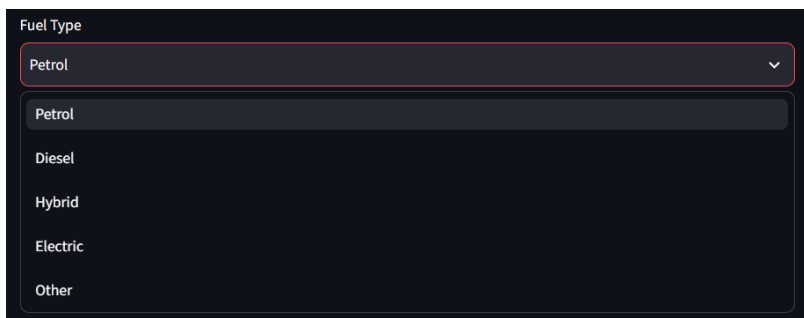
6. (Categorical Input using Dropdowns)

```

54 # Engine Size
55 engineSize = st.number_input(
56     "Engine Size (L)",
57     min_value=0.0,
58     max_value=10.0,
59     value=1.5
60 )
61 # Categorical input fields using dropdowns
62
63 # Transmission type
64 transmission = st.selectbox(
65     "Transmission",
66     ["Automatic", "Manual", "Semi-Auto"]
67 )
68
69 # Fuel type
70 fuelType = st.selectbox(
71     "Fuel Type",
72     ["Petrol", "Diesel", "Hybrid", "Electric", "Other"]
73 )
74 # Input should be feed to the model

```

Output:

7. (Text Input and Predict Button)

```

74 # Text input for the car model name
75 model_name = st.text_input("Car Model Name")
76
77 # Predict button
78 predict_button = st.button("Predict Price")
79 # Input should be feed to the model

```

Output:

Fuel Type

Petrol

Car Model Name

Predict Price

8. (Creating Input DataFrame & Encoding)

```
# Create dataframe
input_df = pd.DataFrame({
    "model": [model_name],
    "year": [year],
    "transmission": [transmission],
    "mileage": [mileage],
    "fuelType": [fuelType],
    "tax": [tax],
    "mpg": [mpg],
    "engineSize": [engineSize]
})

# One Hot Encoding
input_df = pd.get_dummies(input_df)

# Match training columns
input_df = input_df.reindex(columns=encoded_columns, fill_value=0)

# Scale numerical columns
numeric_cols = ["year", "mileage", "tax", "mpg", "engineSize"]

input_df[numeric_cols] = scaler.transform(input_df[numeric_cols])
```

Output:

Car Model

fiesta

Predict Price

Predicted Price: £14,917.19

9. (Feature Scaling and Prediction)

```
# Scale numerical columns
numeric_cols = ["year", "mileage", "tax", "mpg", "engineSize"]

input_df[numeric_cols] = scaler.transform(input_df[numeric_cols])

# Prediction
prediction = model.predict(input_df)[0]

st.success(f"Predicted Price: £{prediction:,.2f}")
except Exception as e:
    st.error(f"Error : {e}")
```

Output:

Car Model

focus

Predict Price

Predicted Price: £14,917.19

10. (Mini Project – Complete Streamlit App)

Ford Car Price Predictor

Enter the car details below to predict its selling price.

Manufacturing Year

2018

Mileage

30000

Road Tax

150

MPG

55.40

Engine Size

1.50

Transmission

Manual

Manual

Fuel Type

Petrol

Car Model

focus

Predict Price

Predicted Price: £14,917.19

Description:

Ford Car Price Prediction Mini Project

This project is developed using Python, Streamlit, Pandas, Joblib, and Scikit-learn. The application takes car details from the user, preprocesses the input

using One-Hot Encoding and StandardScaler, predicts the selling price using the trained Linear Regression model, and displays the result in a user-friendly Streamlit interface.